

# Next-Gen AI Compiler

## Agents On The Control Plane

Agents Own Search · Orchestration · Synthesis.  
Compilers Own Lowering · Legality · Measure · Fallback.

**Ask: Is The Hybrid Prediction (~2027-31) The Right Bet For Your Roadmap?**

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Expert Briefing · ~30-35 min + Q&A · Prediction-first

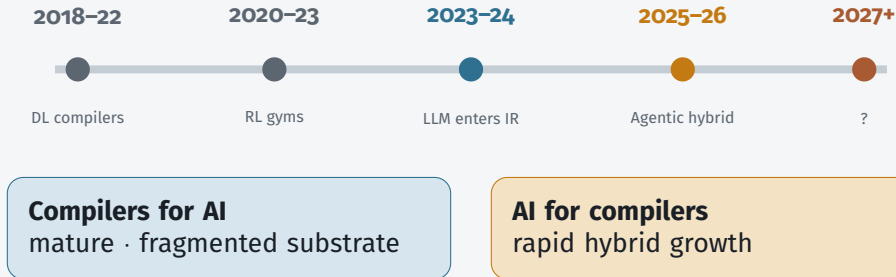
# Agenda

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1. Trends · Keep Substrate
  2. Verdict + Architecture
  3. Roadmap Checkpoints
  4. Commercial Blockers + Gaps
  5. Technical Prediction (Techniques)
  6. Stance · Ask · Appendix

# Trend backdrop — two stacks converging

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Next-gen  $\neq$  throw away LLVM/MLIR — make them **agent-addressable**.

# Six active trends

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## A — Hybrid guidance

Free IR rewrite fragile  
mlirAgent < identity; LLM4IR  
CFG/exec fail on dumped LLVM  
Constrain: hints / ACFs / typed IR  
AgentCompile · HintPilot · Meta LLM Comp. ·  
mlirAgent · LLM4IR

## B — RL → agents

CompilerGym → tool agents  
Heuristics as shippable C++  
Workflow compile · freeze · place  
Compiler-R1/ISCAS · Magellan/DeepMind · Flow-  
Compile/UMass+MIT

## C — MLIR + Triton

PyTorch→Inductor→Triton  
Also StableHLO→XLA/IREE  
Peak often vendor libs / Tile  
Meta · OpenXLA/Google · OpenAI Triton · NVIDIA

## D — Kernel agents

KernelBench: correct *and* faster  
One-shot often <20%; fusion hard  
GEAK v4 e2e · Cake · Argus  
AMD · NVIDIA/CMU · CausalFlow · KernelBench:  
Stanford

## E — Verify in the loop

Unit/golden → numerical  
→ Alive2 / SMT / CBMC  
Weak on GPU races / FP noise  
Alive2 · Argus SMT · T-LLM/CBMC · LLM-VeriOpt

## F — Broader object

Mid-decode diagnosis  
FMware: prompts / agents / knobs  
Agents as compiler engineers  
Compiler.next/Queen's · Magellan/Google ·  
Claude C/Anthropic

# Keep substrate, change control plane

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## Preserve

deterministic lowering  
library kernels  
build-CI regressability  
local formal checks

## Adopt

advisory search  
multi-agent orchestration  
heuristic synthesis  
profile / verifier feedback

**Anti-pattern: replace opt/Inductor  
with unconstrained LLM codegen**

# Executive verdict

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Agents reshape the **control plane**  
more than they replace the **data plane**.

**Control plane:** search / orchestrate / synthesize

**Data plane:** lower / legality / measure / fallback

Compilers for AI  
× AI for compilers

Hybrid LLM-compiler loops

4th job Tier A:  
bring-up / codesign

## Will not happen soon

- Unconstrained LLM replaces Inductor/opt
- Autonomous chip tape-out via compiler agents (C10)

## Four agent jobs — architecture spine

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### (a) Online

propose →  
measure  
admit

CompileIQ / GEAK v4  
Cake / Argus

### (b) Offline

evolve heuristics  
→ C++ / MLGO

Magellan  
AlphaEvolve

### (c) Eng.

oracle-gated  
pull request  
/ change

Archer  
llvm-harness

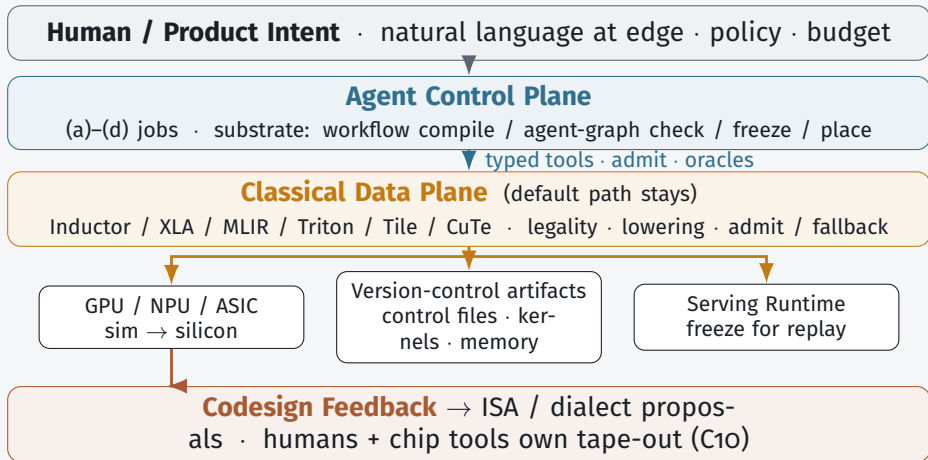
### (d) Bring-up

coverage →  
performance  
sim + silicon

TritorX / KernelEvolve  
Zomboss

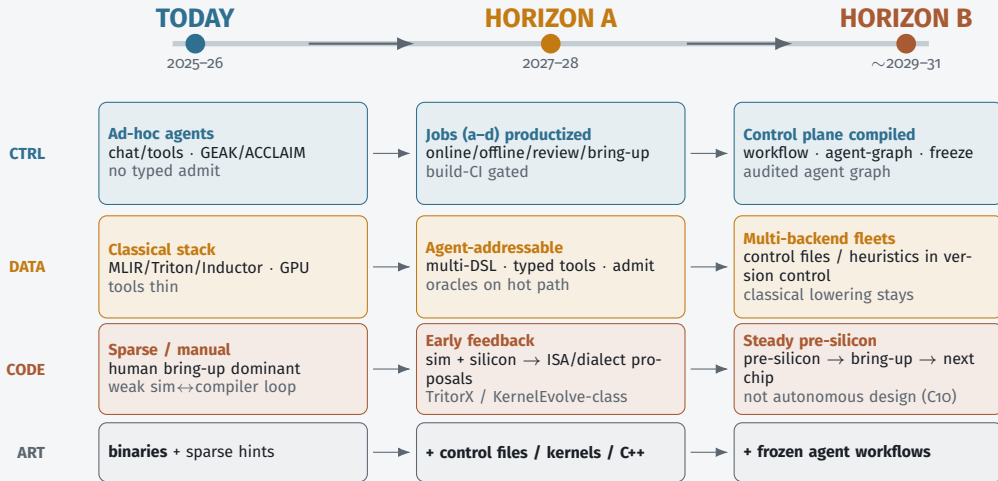
Control plane jobs sit *above* classical lowering — not instead of it.

# Architecture — Target Stack



Invariant: LLM guides search — does not silently define unchecked executable behavior.

# Architecture evolution — component changes



Components thicken left→right; data plane stays; agent graph becomes compiled & amortized.

# Data plane — ~6–7 abstraction bands (not one)

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## Must-have today (L1–L6)

|    |                |                             |
|----|----------------|-----------------------------|
| L1 | Framework      | capture · dynamism          |
| L2 | Portable graph | StableHLO-class · shard and |
| L3 | Mid-IR         | MLIR · layout · passes      |
| L4 | Kernel DSL     | Triton/TileLang/CuTe/TIRx   |
| L5 | Backend-ISA    | PTX · bring-up              |
| L6 | Runtime-serve  | CUDA Graphs · KV            |

Lo\* CPU/LLVM optional · agents sit *above* bands via tools

### L7\* Fleet / cluster

maturing: place + collectives  
today often L2–L3 + runtime

### Not a new IR band

power/energy → objective + oracle  
LLM-oriented IR → agent *face*  
\$/token · safety → policy / admit

### Lean (A6/S6)

keep bands; agents unify *contracts*,  
not one mega-IR

# One universal cost model? — No (Horizon A)

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## Why not one model

Different legality / oracles per band

Choosing the level *is* the product

Cost models stay *local* (MLGO, An-sor)

Future passes need new measured labels

## If bands do not consolidate

Agent compile schema (admit hash)

Typed tools / MCP-class servers

Dialect + oracle + objective plugins

Placement / fleet plugins (L7)

**Size:** local advisors often KB-MB · cross-band *proposer*  $\approx 7B-70B+$  IR LLM (prior only)

Bottleneck = labeled (program, action, HW, energy, SLO) tuples — not parameter count.

**Bet:** small per-band priors + e2e controller under  $F$  — not one mega-cost-model for L1-L7.

# E2E-optimal-seeking — reshape around $F$

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**Product fitness  $F$**  (latency · energy · \$/token · quality · cluster) — Pareto/constrained, not  $\sum$  local costs

**E2E search controller** — joint/bilevel policy · credit across bands · budget/stop/freeze

**L2–L3**

**L4–L5**

**L6**

**L7\***

bands = legality / lower surfaces · local cost = proposal prior only

**Physical  $F$ -admit:** serving A/B (GEAK v4) → pinned traces → energy/fleet · freeze or classical fallback

**Claimed**

e2e-optimal-seeking control plane  
joint search +  $F$ -admit + freeze

**Not claimed**

unique closed-form global optimum  
complexity bound  $\neq$  stop trying

# Merge $\neq$ replace — three levels

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## M1 Soft merge (Horizon A $\rightarrow$ B)

e2e controller + admit/trace over band tools · data plane stays · agents do *not* replace compiler

## M2 Search-surface merge (contested / partial $\sim$ 2029–31)

fewer agent training IRs (C4) · lowers still multi-band · still hybrid

## M3 Hard replace — *not* claimed thru Horizon B ( $\sim$ 2031)

default path with *no* classical admit/fallback · only if C6-A settles

**Perception trap:** “I only talk to an agent” is often M1+freeze (classical lower still ran in CI).  
**Lean (A8 / C6-B):** merge the *optimizer* under *F*; do not delete the execution substrate.

# What ships / does not by 2028

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## Predicted to ship

- Agent-addressable tool interfaces
- Hot-path specialize (not silent default-all)
- Magellan *and* MLGO both live
- Oracle pull-request review in serious orgs
- Coverage-first ASIC bring-up
- Triton-family primary; Cake IR / Argus DSL rising

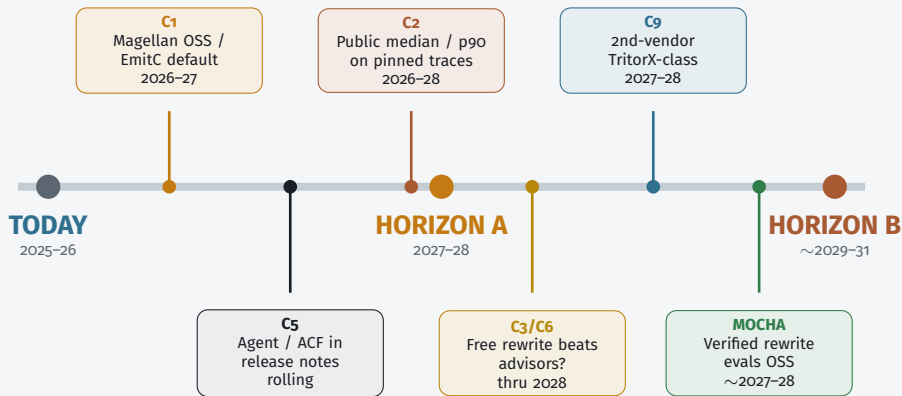
## Does not ship

- Unconstrained LLM replaces opt/Inductor
- One agent IR for all vendors
- Kernel agents uniformly beat eager on fusion ladders
- Autonomous microarch tape-out

**Horizon A success = build-CI gated specialize + oracles + freeze artifacts**

# Roadmap Checkpoints — when the prediction changes

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Falsifiable milestones — watch settlement signals; do not average disagreements.

# Checkpoint C1 — heuristics vs neural advisors

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## A — Magellan path

evolve shippable C++ heuristics

Evolve shippable C++  
(EVOLVE-BLOCK)

Offline coding agent *is* the  
control-plane output

Product = evolutionary coding agent +  
review

## B — MLGO / EmitC path

MLGO = learned in-tree LLVM advisors

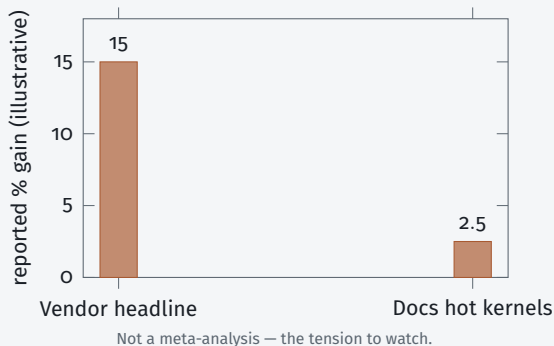
NN advisors stay in-tree

June 2026 PoR: inliner → Android/Fuchsia → Chrome

Agents train/feature NNs

**Settlement:** public Magellan llvm patches that displace MLGO or EmitC-MLGO becomes customer default  
Watch LLVM syncs + OpenEvolve recipes quarterly through 2027

## Checkpoints C2 + C5 — gains & default path



**C2** “Agents become default” needs **median (p50) build-CI wins**, not best-kernel blogs.

p50 / p90 = 50th / 90th percentile. GEAK v4 A/B + Argus peak = *pressure*, not settlement.

**C5** Online specialize (control files / hints) vs offline eng (Magellan/MLGO) — both may live; *which is the default flag?*

**Settlement:** pinned public traces + release notes listing agent / control-file workflows.

# Checkpoints C<sub>3</sub> / C<sub>6</sub> / C<sub>9</sub> / C<sub>10</sub> — interface width & scope

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## C<sub>3</sub> — free rewrite vs advisory

Flip: shared suite where free rewrite consistently wins

⇒ wide rewrite API vs ACFs / hints / Cake IR / Argus / intent (LLM4IR: dump fails)

## C<sub>6</sub> — replace vs control plane

Flip: production *default* path with no classical admit / fallback

⇒ agents replaced the compiler (survey rejects; hold hybrid)

## C<sub>9</sub> — coverage vs peak

Flip: TritonX → KernelEvolve ladder becomes a public playbook

⇒ SKU: coverage SLA then perf; Zomboss = academic analog, not 2nd vendor

## C<sub>10</sub> — codesign vs auto tape-out

Flip: microarch primarily agent-proposed *and* compiler-oracle validated

⇒ enters chip-design tooling (today: agents stress kernels/IR only)

# Commercial blockers — top 5

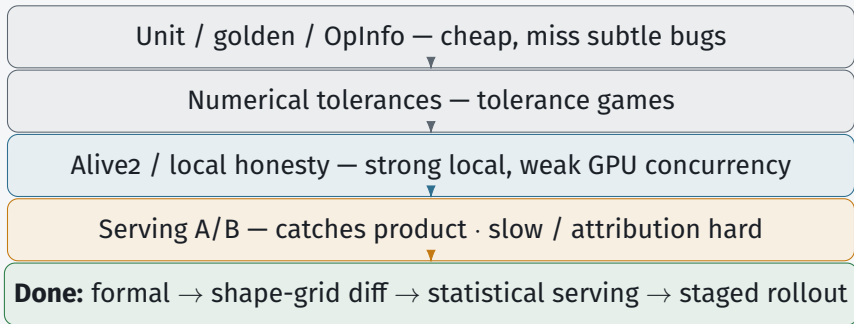
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- 1 Oracles strong enough for money  
Fast-but-wrong · liability
- 2 Cost / replay / when agents run  
Nondeterministic \$N compiles
- 3 Agent $\leftrightarrow$ compiler contract  
Natural-language-only = demo
- 4 Distributional production evidence  
No default-path A/B
- 5 Ownership · supply chain · human review  
Unowned agent code in trusted base

Each blocker gets one slide — productization gaps, not research wishlist.

## Blocker 1 — Oracles for money

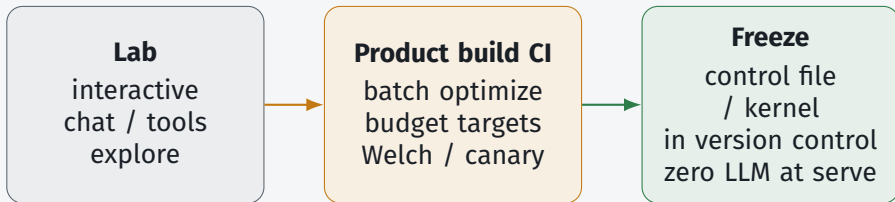
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**Room question:** who owns false negatives when admit passes and production miscompiles?

## Blocker 2 — Cost, replay, when may the agent run?

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Cache key: (IR hash, hardware, compiler ver, agent policy) · budget \$/%gain

Falsifies commercially: “chat with compiler on every build” without spend/latency targets.

## Blocker 3 — Agent $\leftrightarrow$ compiler contract

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**A. Natural language + pasted logs** — demo only

**B. Structured admit traces** — build CI / bill of materials

**C. Typed tool interfaces** — required action space

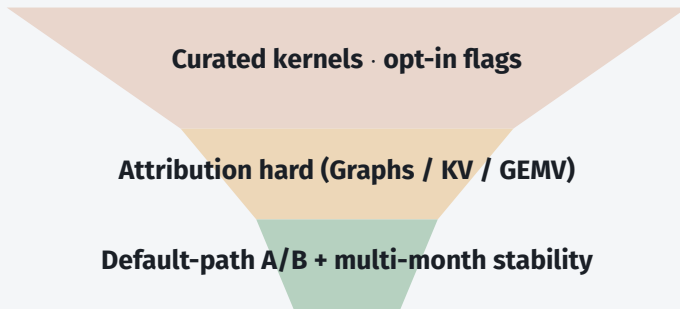
**D. Hybrid** — natural-language view over traces

```
admit_record = {graph_hash, hw_id, compiler_ver, action[],  
oracle[], artifact_digest, policy_id}
```

Lean: **C** + **B**; Natural language is a view, not  
source of truth. ACF = portable compiler knobs.

## Blocker 4 — Distributional production evidence

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Marketing bar: median + 90th-percentile gains + cost-to-compile  
· persistent serving throughput, not single-kernel headlines

## Blocker 5 — Ownership, security, human review

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### Trusted-base risk

agent kernels /  
heuristics enter  
trusted base

### Ownership

CODEOWNERS  
signed provenance  
sandbox

### Human-review capacity

agents  $\uparrow$  drafts  
reviewers =  
bottleneck

Lean: human CODEOWNER + signed admit + sand-  
box · oracle auto-merge only for **narrow** action classes  
Generic forge AI / coding-agent harnesses  $\neq$  compiler-oracle review (C7)

# Gap map — what blocks the prediction, not a wishlist

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## HIGH

4.1 prod evidence

4.2 correctness

4.3 cost/replay

4.4 cross-stack

4.10 benchmarks

## MED-HIGH

4.5 hardware-native interfaces

4.7 training data

## MEDIUM

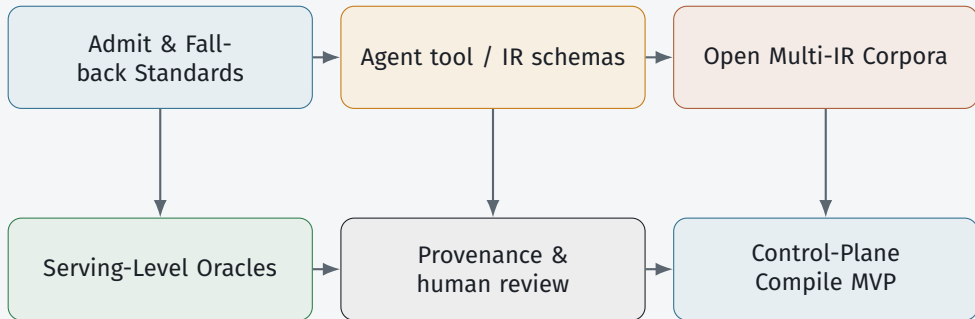
4.6 workflow compile

4.8 human-review process

4.9 security

# Cross-Cutting Research Agenda

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Near-term research themes → next: technique map T1-T10 (exists / missing / unlocks).

# Technical Prediction — Accelerate The Roadmap

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To settle checkpoints and ship Horizon A,  
enhance techniques *inside and outside* the compiler (T<sub>1</sub>–T<sub>10</sub>).

## Within (T<sub>1</sub>–T<sub>5</sub>)

- T<sub>1</sub> Typed agent↔compiler interfaces
- T<sub>2</sub> Admit / fallback machinery
- T<sub>3</sub> Control files, hints, fingerprints + replay
- T<sub>4</sub> Heuristic hooks & in-tree advisors
- T<sub>5</sub> Dialect / ISA feedback sinks

## Outside (T<sub>6</sub>–T<sub>10</sub>)

- T<sub>6</sub> Serving-level oracles & production A/B
- T<sub>7</sub> Open multi-IR corpora (+ negatives)
- T<sub>8</sub> Unified benchmark ladder
- T<sub>9</sub> Provenance, ownership, HITL process
- T<sub>10</sub> Agent-workflow compile / freeze / place

Enhancing only opt/Inductor/Triton internals is not enough. Next: exists · missing today · checkpoint unlocks.

# Technical Prediction — Within The Compiler (T<sub>I</sub>–T<sub>5</sub>)

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Exists · Missing · Unlocks

## T<sub>1</sub> Typed interfaces

→ C<sub>3</sub>, C<sub>5</sub>, C<sub>6</sub>

Exists: CompileIQ · Cake IR · Argus · TIRx · TileLang LSP · LLM4IR (neg.)

Missing: portable schemas across MLIR · Triton/Gluon/TLX · Tile · Cake · Argus · TIRx

## T<sub>2</sub> Admit / fallback

→ C<sub>6</sub> hybrid

Exists: Archer · Cake gates · Argus SMT · FlashInfer-Bench

Missing: shared admit *product* + trusted deterministic fallback

## T<sub>3</sub> Control files + replay

→ C<sub>2</sub>, C<sub>5</sub>

Exists: CompileIQ ACFs · FlashInfer Trace + `apply()`

Missing: content-addressed keys; golden replay on model upgrade

## T<sub>4</sub> Heuristic hooks / advisors

→ C<sub>1</sub>

Exists: Magellan / AlphaEvolve · MLGO · EmitC PoR

Missing: settled Magellan vs MLGO *default* on named apps

## T<sub>5</sub> Dialect / ISA sinks

→ C<sub>9</sub>, C<sub>10</sub>

Exists: TritonX · KernelEvolve · Cake IR evol · Zomboss

Missing: first-class change-proposal surfaces (not auto tape-out)

# Technical Prediction — Outside The Compiler (T6–T10)

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Exists · Missing · **Unlocks**

## **T6 Serving oracles / A/B**

→ **C2**

Exists: Alive2 · **FlashInfer-Bench**+`apply()` · **GEAK v4** A/B

Missing: whole-program / GPU-race / FP; multi-month default-path A/B

## **T7 Multi-IR corpora**

→ **Selectors**

Exists: Meta LLM Compiler · **ComPile** · **KernelBook**→TritonRL · DRTriton

Missing: versioned MLIR / Tile / StableHLO + failed / miscompile negatives

## **T8 Benchmark ladder**

→ **C2, C9**

Exists: KernelBench(-X) · FlashInfer-Bench · llvm-bench

Missing: full IR→kernel→fused→serving + cost-to-compile

## **T9 Provenance / HITL**

→ **C7**

Exists: Magellan C++ · Archer · llvm-harness

Missing: CODEOWNERS + signed admit records + sandbox for proposals

## **T10 Workflow compile / freeze**

→ **Horiz. B**

Exists: FlowCompile · Auto · AgentFlow · DSH runtime

Missing: shared agent-graph IR + fail-closed CI → Horizon B

# Technical Prediction — What Each Checkpoint Needs

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**C1**  $\leftarrow$  **T4**

Magellan-class synthesis or EmitC-MLGO customer default on named apps

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**C2**  $\leftarrow$  **T3 + T6 + T8**

Replayable artifacts + serving oracles + ladder

FlashInfer-Bench + GEAK v4 A/B = pressure, not settlement

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**C5**  $\leftarrow$  **T1 + T3**

Typed interfaces + freeze-before-serve named in release notes

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**C3 / C6**  $\leftarrow$  **T1 + T2**

Constrained tools + admit/fallback (lean: hybrid advisory)

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**C9**  $\leftarrow$  **T5 + T8**

Coverage  $\rightarrow$  perf + dialect sinks; Zomboss analog  $\neq$  2nd-vendor shipping

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**C10**  $\leftarrow$  **T5**

ISA/dialect *proposals only*; humans + chip-design tools own tape-out

# Technical Prediction — Critical Missing Parts Now

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## Highest-leverage bets to accelerate Horizon A

### 1. Money-grade oracle stack (T2+T6)

w/o → demos

local formal → shape-grid diff → serving statistics → staged rollout

### 2. Replayable artifact contract (T3)

w/o → CI rejects

control files / kernels / heuristics with cache keys + golden replay

### 3. Portable agent compile interface (T1)

w/o → re-glue

LLM-oriented IR: summaries · intent · typed surface · admit — *not* a new L-band

### 4. Open ladder + multi-IR data (T7+T8)

w/o → incomparable

correctness × speed × \$/compile + negative (failed) examples

Survey lean: T1–T5 ship as product surfaces; T6–T10  
equally first-class — mostly outside classical lowering.

# Org Adoption Questions

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1. Online (CompileIQ) vs Offline (Magellan) — which matches your release model?

2. Oracle stack: Alive2 / golden / serving A/B — who owns false negatives?

3. Agent-visible IR: dump vs summaries / Cake / Argus / TIRx / Gluon?

4. How do you cache/regress traces across compiler *and* model upgrades?

5. Max \$/build for a median X% win? Named maintainer per admitted artifact?

# Working stance until conflicts settle

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- 1 Hybrid: agents search; compilers lower (not replace)
- 2 Magellan || MLGO — parallel bets (C1)
- 3 Discount single-number speedups (C2)
- 4 Constrained actions + strong oracles (C3)
- 5 Demote generic SCM / coding-agent AI (C7)
- 6 Coverage→performance; no autonomous chip design (C9/C10)
- 7 ~6–7 data-plane bands; plugins if not consolidated (A6/S6)
- 8 E2E-optimal-seeking: joint search under  $F$  (A7/S7)
- 9 Soft merge M1 by Horizon B; hard replace M3 not claimed (A8/C6)

# Commercial checklist — handout

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## Architecture

typed tools + admit  
traces  
version control >>  
dense memory >>  
scratchpad  
state-machine plan for  
service targets  
freeze before default-  
on

## Trust

layered oracles  
CODEOWNERS + prove-  
nance  
joint version pins  
A/B before “prod de-  
fault”

## Business

sell regressable arti-  
facts  
+ build-CI quota  
customer owns out-  
puts  
coverage then perfor-  
mance service target  
token budget per  
%gain

# Discussion

Thank you — hybrid bet stays the ask

1. Which checkpoint flips your investment first — C1, C2, or C9?
2. Harder commercial bar: **oracle strength** or **replay/freeze**?
3. Building job (a), (b), or (d) first — what do you refuse online?

Ask again: is the hybrid bet right for your roadmap? Watch settlement signals — do not average disagreements.

# Appendix

## Evidence Store Snapshot

Publications digests · commercial product signals · open repositories.  
Tiered for the agentic-compiler prediction — not a full catalog.

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## Appendix — evidence map

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### Publications

136 digests  
one file per source  
papers · blogs · talks  
forge pages · minutes  
★ = prediction-critical

### Products

Commercial SKUs as  
*prediction signals*  
Tier A shapes the future  
Tier B = data-plane de-  
faults  
not a full SKU list

### Repositories

GitHub / Gerrit / forge  
artifacts tiered A/B/C  
A = agents reshape  
compile  
C = generic forge AI only  
not an exhaustive forge  
map

Use ★ digests + Tier A products/repos when a checkpoint or blocker is contested.

## Appendix — Tier A commercial signals

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Company offerings that shape what ships — agent loop / kernel surface / cloud agent

|                          |                                                                                 |
|--------------------------|---------------------------------------------------------------------------------|
| <b>Google / DeepMind</b> | AlphaEvolve Cloud (GA); Magellan + MLGO (heuristics <i>and</i> advisors)        |
| <b>NVIDIA</b>            | CompileIQ (+ skills); Tile IR; Cake IR (research); TRT-LLM agents               |
| <b>AMD</b>               | GEAK v4 e2e A/B (sglang/vLLM warm path)                                         |
| <b>CausalFlow et al.</b> | Argus SMT / tags (MI300X research — not an AMD SKU)                             |
| <b>Meta</b>              | LLM Compiler / KernelLLM; TritorX + KernelEvolve; Helion / TLX                  |
| <b>FlashInfer</b>        | FlashInfer-Bench (serving-trace ladder + <code>apply()</code> into SGLang/vLLM) |

Tier B baselines: TensorRT-LLM · Inductor · XLA/StableHLO · FlashInfer · Modular  
MAX · OpenVINO · Neuron · Hexagon-MLIR

# Appendix — Tier A open repositories

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## Offline / review / heuristics

OpenEvolve · HeuriGym · Archer · llvm-harness · Compiler-R1 · mlirAgent

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## Online / kernels / bring-up

ACCLAIM · CompileIQ · GEAK v4 · KernelAgent · KernelBench(-X)  
FlashInfer-Bench · AutoKernel · Helion · TritonX / KernelEvolve / Zomboss

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Tiers: A = agents + domain oracles change heuristics/kernels/knobs/review

B = data-plane hosts · C = generic forge AI (demoted for prediction)

Not A: DeepSeek Harness = agent runtime (plugins / session log), not compiler oracles (C7)

# Appendix — prediction-critical digests (★)

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Highest-signal sources — mechanisms, not headlines

## Vision / funded

Compiler 2.0 Ken Kennedy (MIT) · DARPA MOCHA / Aarno

## Offline / online

Magellan · ACCLAIM · EmitC-MLGO · LLM4IR ★

## Bring-up / codesign

TritorX · KernelEvolve · Zomboss ★ · KForge

## Kernels / ladder / data

Cake ★ · Argus ★ · GEAK v4 ★ · FlashInfer-Bench ★ · KernelBook

Full digest index 136 entries. Technique map: SURVEY §5.8 T1–T10.

# Appendix — publication groups

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**30** GPU kernels & inference   **15** Agentic & RL   **14** Source control & review  
**12** Classic DL compilers   **10** Company infra   **10** Forums & workshops  
**8** Surveys & vision   **12** Foundation LLMs   **6** MLGO & RL gyms  
**6** HW codesign & bring-up   **6** Commercial products   **6** Control-plane substrate   **1** Correctness lineage

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## How digests are used

Mechanism first; demote generic forge AI; keep both sides — do not average  
Update prediction only when Tier A / ★ evidence moves